Figure 6). The model with the parallel stacking scheme, therefore, cannot explain the ^{13}C NMR data.

To get an impression of how different the inverted and parallel spectra look, we show both the parallel and inverted g_S spectra in Figure 15b. The line shape of each carbon or nitrogen class has its "natural" line width (the standard deviation of the chemical shifts belonging to the respective class with respect to their average, so $\gamma = 1$). We choose a generic shift-shielding relation with $a = 1$, and $b = 174.6$ ppm together with a generic amplitude of $A = 1$ to plot the spectra. We also want to draw attention to the asymmetry of the isolated methyl resonance: while this resonance manifests as asymmetrical in the experimental spectrum, the parallel model produces a symmetric resonance. From all these considerations, and the fact that the inverted spectrum is in near-perfect agreement with experiment, it is clear that the parallel stacking configuration, if present, is negligible with respect to the inverted stacking configuration. This case shows that NMR could also be used to determine stacking configurations in COFs.

4. CONCLUSIONS

In this paper, we propose three approaches to construct computational NMR spectra for COFs through the usage of quantum mechanical static structural models. All models start from the same structural model and the chemical shielding data but approach it in a different way, ranging from no averaging over shieldings to maximal averaging. It is clear that all three approaches could be deemed viable alternatives for the production of a model carbon spectrum, as each approach produces a spectrum that is in decent agreement with the experimental spectra. In reality, there is mobility of the framework, and each carbon site is dynamic. What we should aim to mimic is an ensemble average over configurations that is representative of the dynamic COF structure. During the collection of the NMR signal, carbon sites propagate through multiple possible geometries, and therefore, the actual NMR resonance that a single carbon site produces is an average resonance. One could resort to MD for the propagation of the system either at the DFT level of theory or using a force field. The first, however, is expensive for systems of this size. The computational cost could still be a problem in the dynamic approach with force fields because the force field still has to be derived from more expensive calculations, which is in itself not always easy in complex materials. In any case, given the fact that COFs can be placed in different stacking configurations and with different orientations for the linkers, it can become impractical to perform MD runs for all possible configurations, whereas it might still be feasible to use static calculations for all candidate structures. Within this work, we have shown that a static approach can yield a great deal of information without the need to perform explicit MD simulations. Instead, the inherent spatiotemporal averaging of an experimental measurement can be mimicked by an appropriate configurational averaging technique for the chemical shifts, grounded in a particular grouping strategy of atomic sites. The three approaches g_I, g_S, g_R each have their own specific merit and downsides:

- The uniform model (g_I) uses a single line shape per atomic site. Since we are using a static model, it is therefore not very physical that each carbon site produces a single line shape with one specific chemical shift. In this sense, the model is not a good approximation for the ensemble average that we aim to represent. However, the spectrum that it produces can visually still be similar to the experimental spectrum, therefore it could serve as a quick first check that requires little postprocessing.
- The symmetrically inequivalent average model (g_R) is, in a sense, the opposite of the g_I model, because it averages out a lot of information where the g_I model does not average anything. The atomic sites are grouped based on the symmetry of the nodes and linkers. However, the symmetry of the building block inside the COF material can be lower than the isolated molecule. When the chemical shift is overly averaged, information can be lost, as carbon sites that are not equivalent inside the framework are treated as being equivalent.
- The SBU average model (g_S) is an intermediate between the g_I and the g_R model, as it still uses symmetry to group the carbon sites into inequivalent classes. However, in this model there is more freedom, as the classes of the g_R model are partitioned. Each atom in the node or linker represents an inequivalent class, and the average is taken over the nodes/linkers present in the system. With this model, we approximate the ensemble average the closest, without the need to perform expensive MD simulations at the DFT level of theory.

Furthermore, we hypothesize that, for some systems, there exist better choices for the g_S model classes than just basing the classes on the symmetry of the underlying building blocks. For example in the case of COFs, if there is reason to believe that carbon atoms across different layers will produce a separate resonance in the NMR spectrum, it will be better not to average the chemical shifts over these different building blocks across layers. Rather, in that case, it would be best to simply increase the number of classes (and, if necessary to obtain good statistics, increase the unit cell size) and only average the chemical shifts over classes within the same layer of COF material. However, increasing the number of classes further in this system is not beneficial in this case, as the number of carbon atoms per class in such a modified g_S model becomes too small to retain decent statistics in this case. The principle of increasing the number of carbon classes to limit the averaging of chemical shifts to carbon atoms residing within the same layer can also be applied to the g_R model, the resulting spectrum of which may be found in Section S5. Looking beyond the materials discussed in this work, an optimal choice for the different classes in the g_S model hinges on the (in)equivalence of the atomic sites within the material, originating from the invariant structural features of the material.

Similar to systems where significant geometrical differences are expected to be present across layers, in cases in which some form of turbostratic disorder is present, multiple regions of the COF can have different space groups. In that case, some carbon classes that by default in the g_S model are treated to be equivalent are better kept as separate classes. In a case such as COF-5,[55] it is possible that considering the different existing symmetries of the (different regions of the) framework is preferable over defining the carbon classes based on the symmetry of the building block.

We also showed that the assignments of carbon spectra that are routinely performed by experimentalists are not always fully

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